

Synthesizer

Seungwoo Ji

Written Description of Operation

This project is a simple synthesizer that creates and changes sound using basic electronics.

The VCO is where the sound starts. It makes different types of sound waves, like sawtooth waves (which look like sharp, jagged lines) and pulse waves (which look like blocks and the pulses can be adjusted with a potentiometer). These waves are created using a special chip called a CD40106 Schmitt trigger and other small parts like diodes, capacitors, and resistors. I

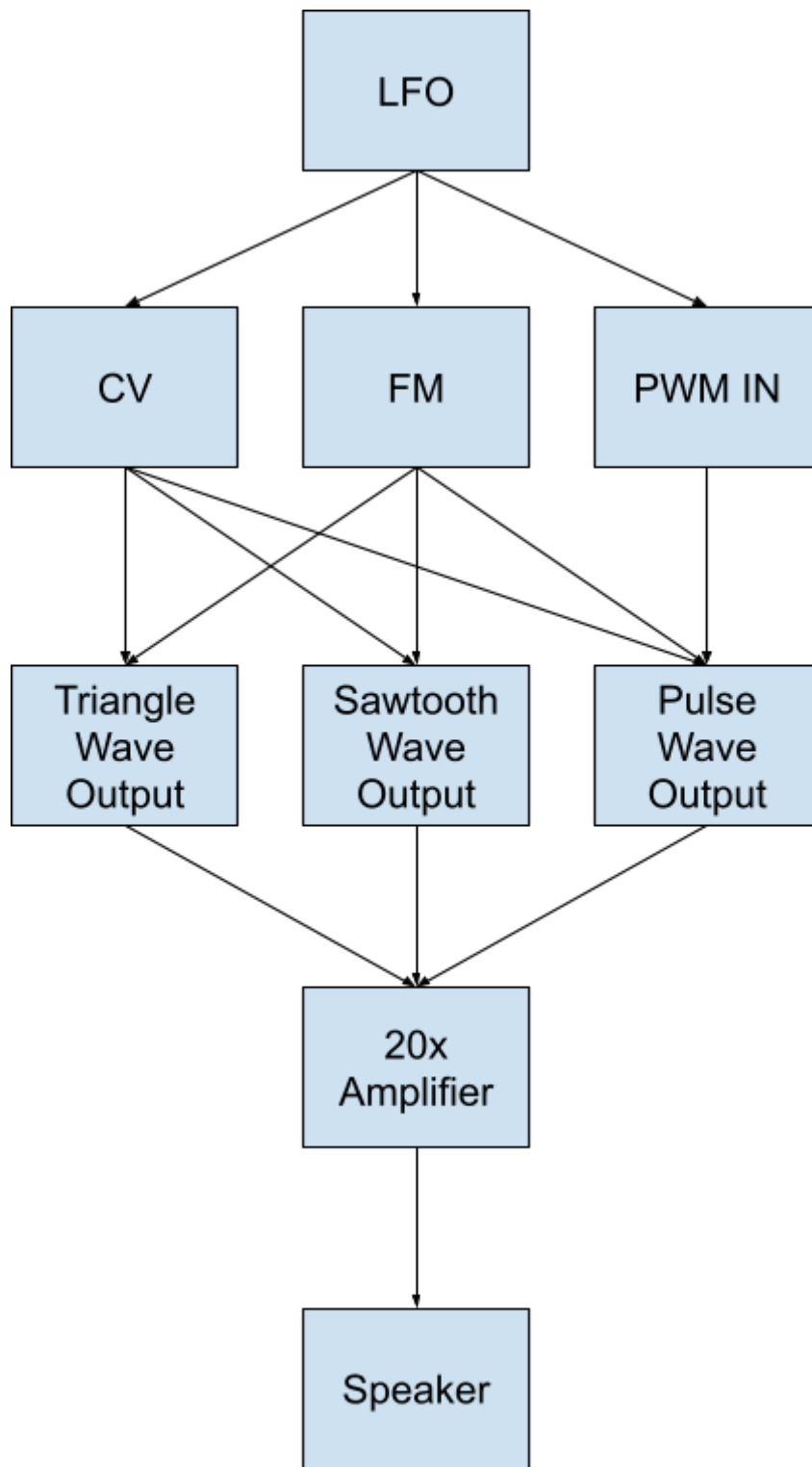
The sawtooth wave made by the VCO is turned into a triangle wave using another circuit. This makes the sound softer(?) or just different. Those are the three wave outputs used to create the sound.

The LFO then creates slow waves that can be used to change the waves that were created by the VCO. For example, it can make the pitch go up and down, which is called vibrato, or change the volume to make a tremolo effect.

After the sound is shaped, it goes through an amplifier to make it louder. I can change the input wave to the amplifier between the three waves to play different sounds. Finally, the amplified wave plays through a speaker so you can hear it.

In conclusion, the system works by starting with simple sound waves, changing them in different ways with circuits, and then sending the final sound out to a speaker.

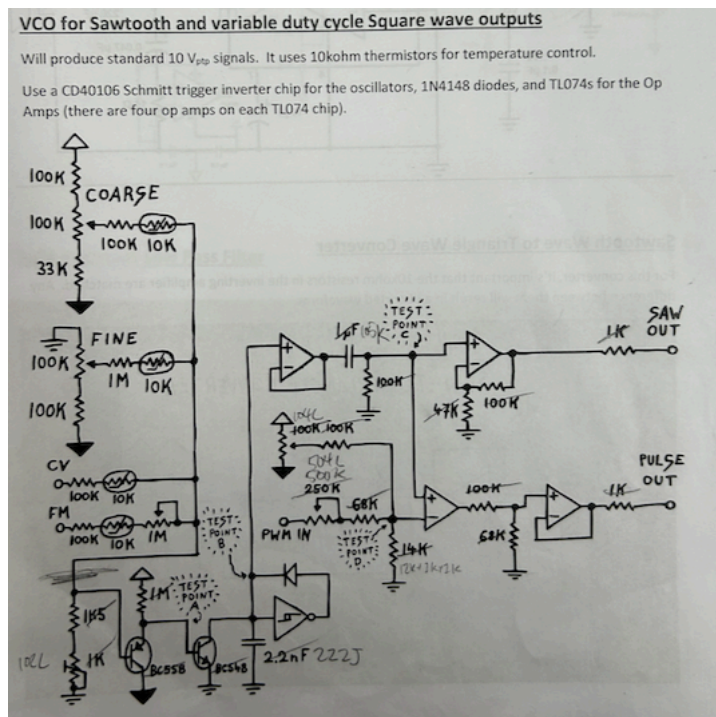
Block Diagram



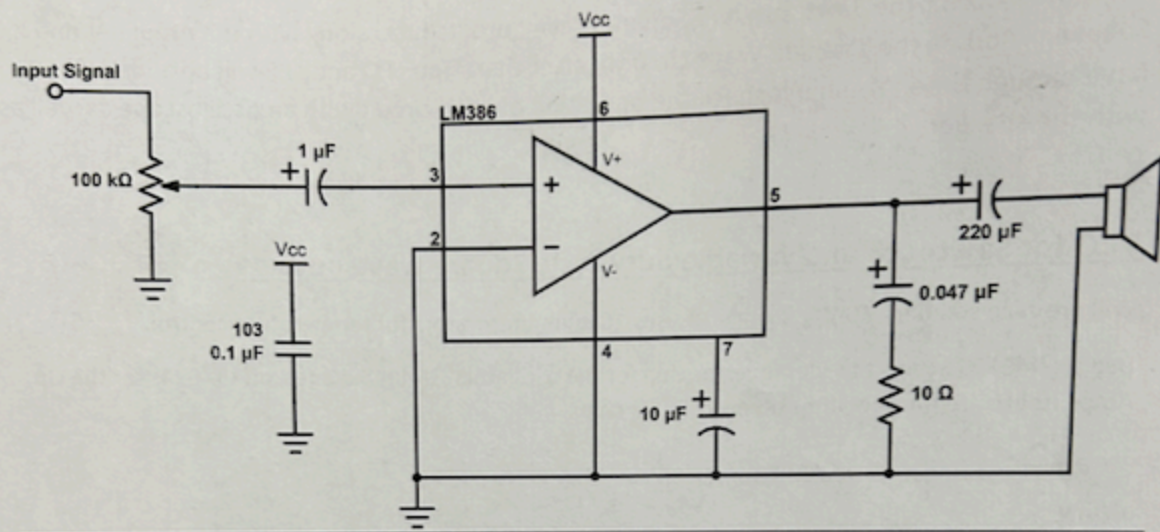
Similarities and Differences

My project is very similar to the device it was modeling. Mr. Kirk provided me with the original device schematics and data sheets. The only differences are battery voltage and the fact that my project does not include a low pass filter which would just affect the final sound output by allowing low frequencies to pass through.

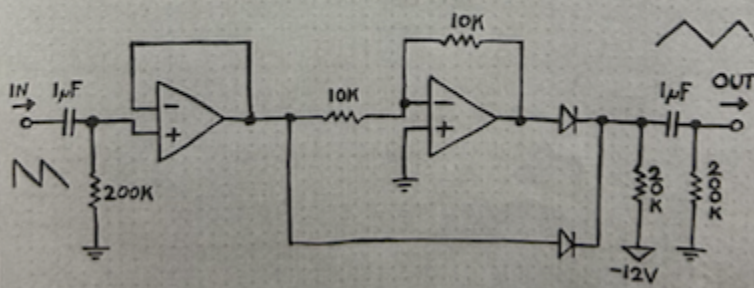
Original Schematics

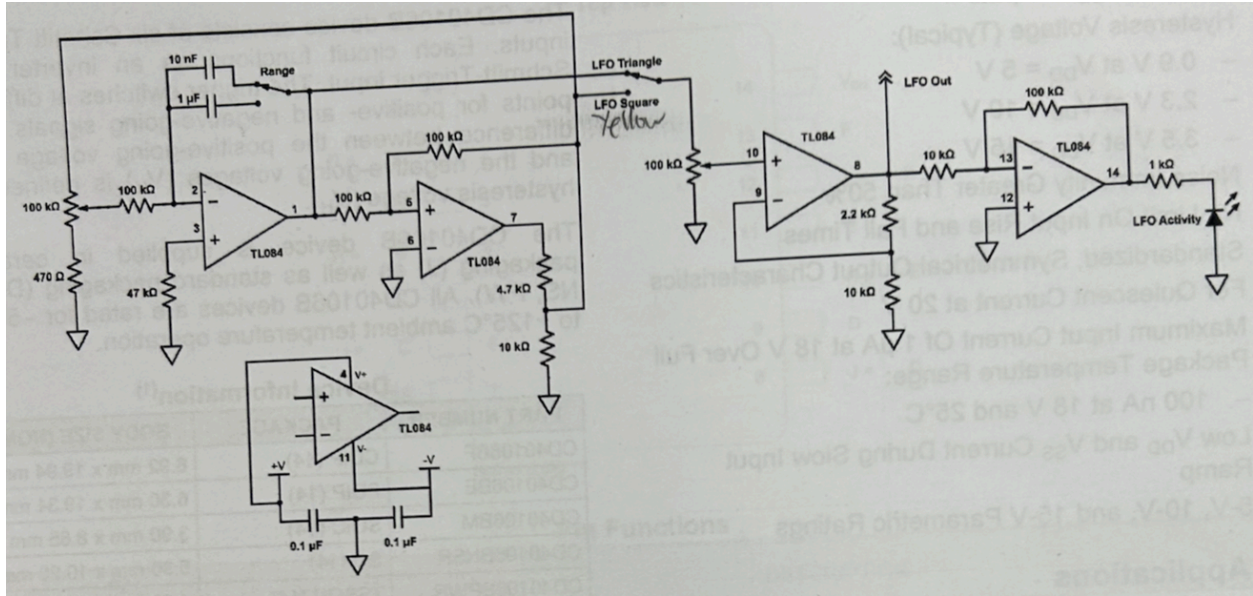


20x Amplifier Circuit

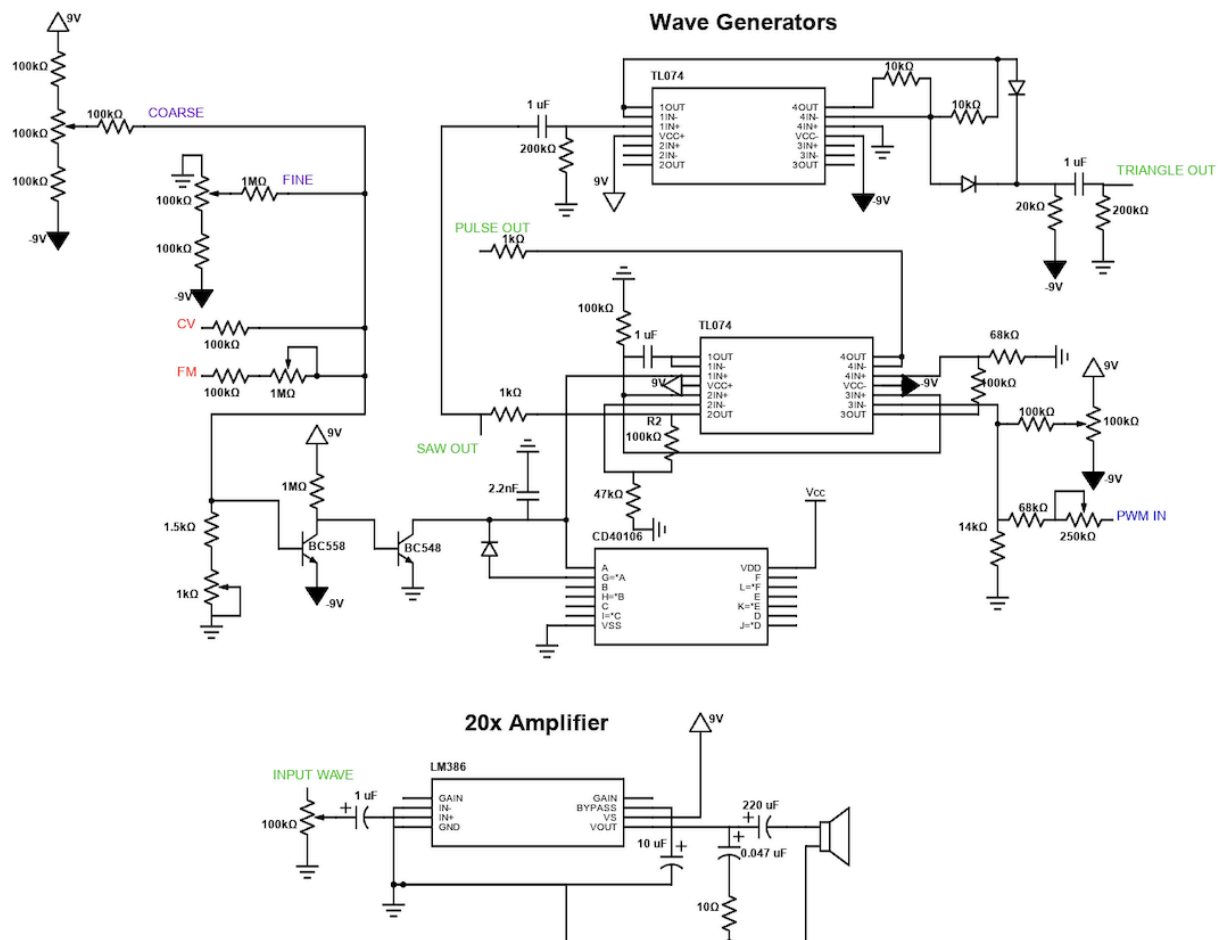


SAW-TO-TRIANGLE CONVERTER

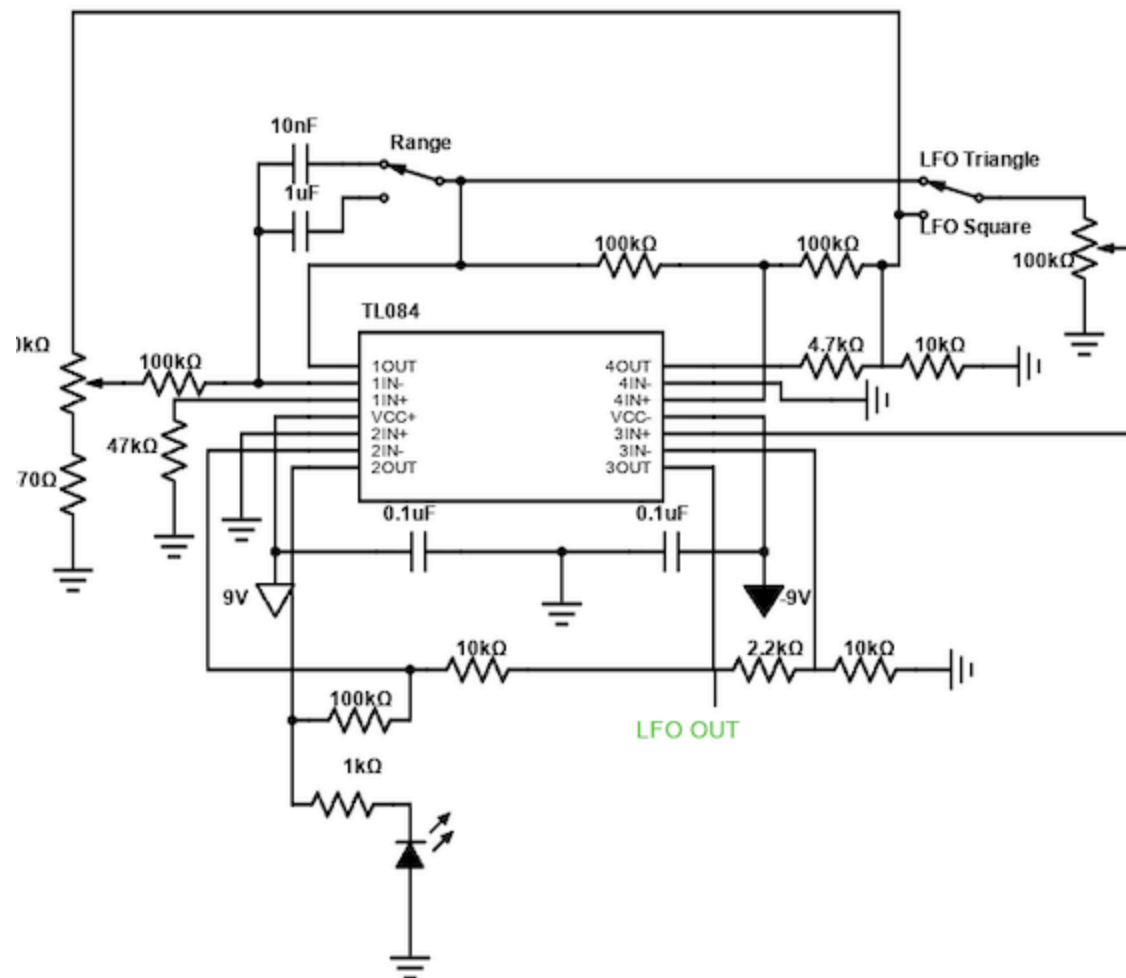




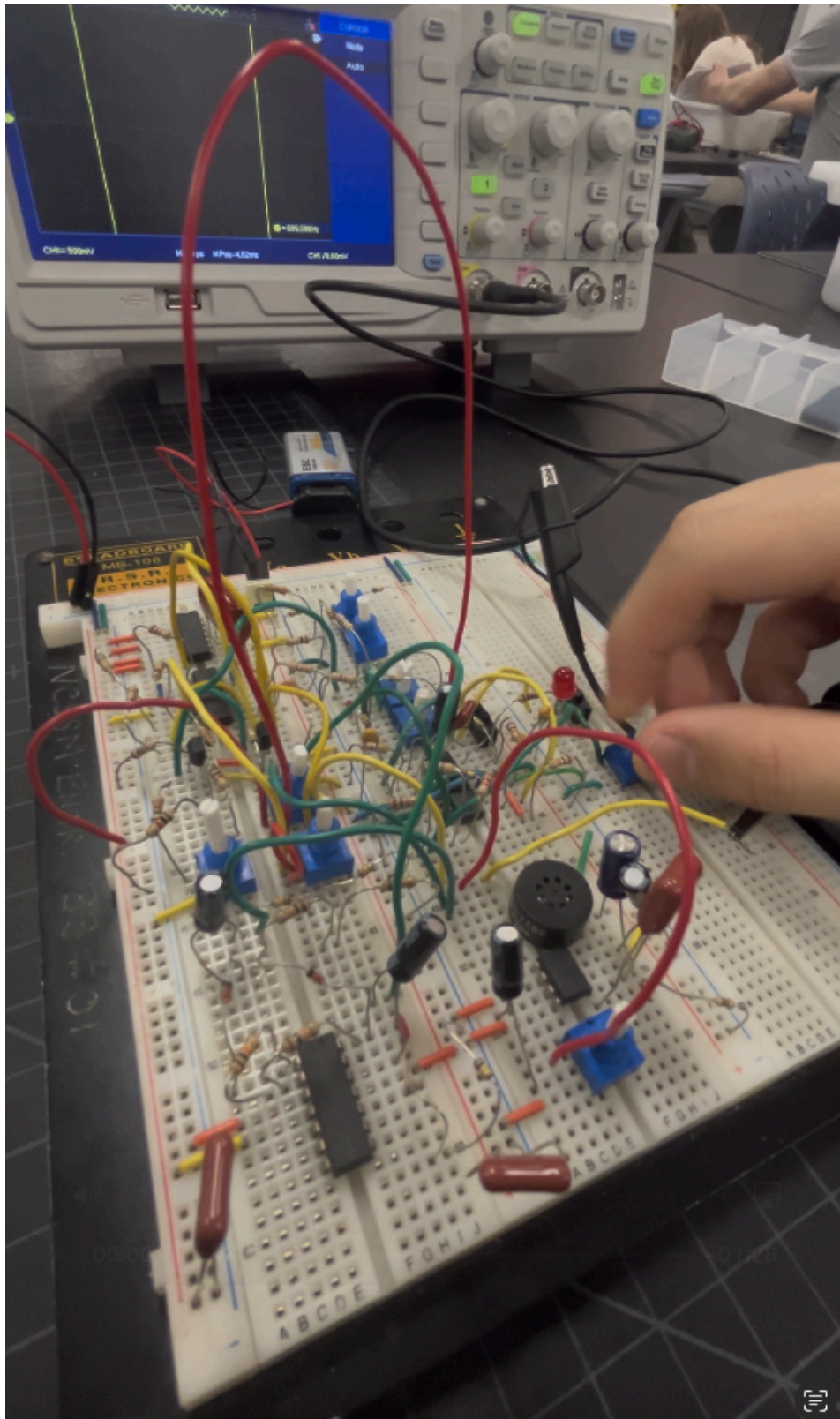
Wiring Schematic

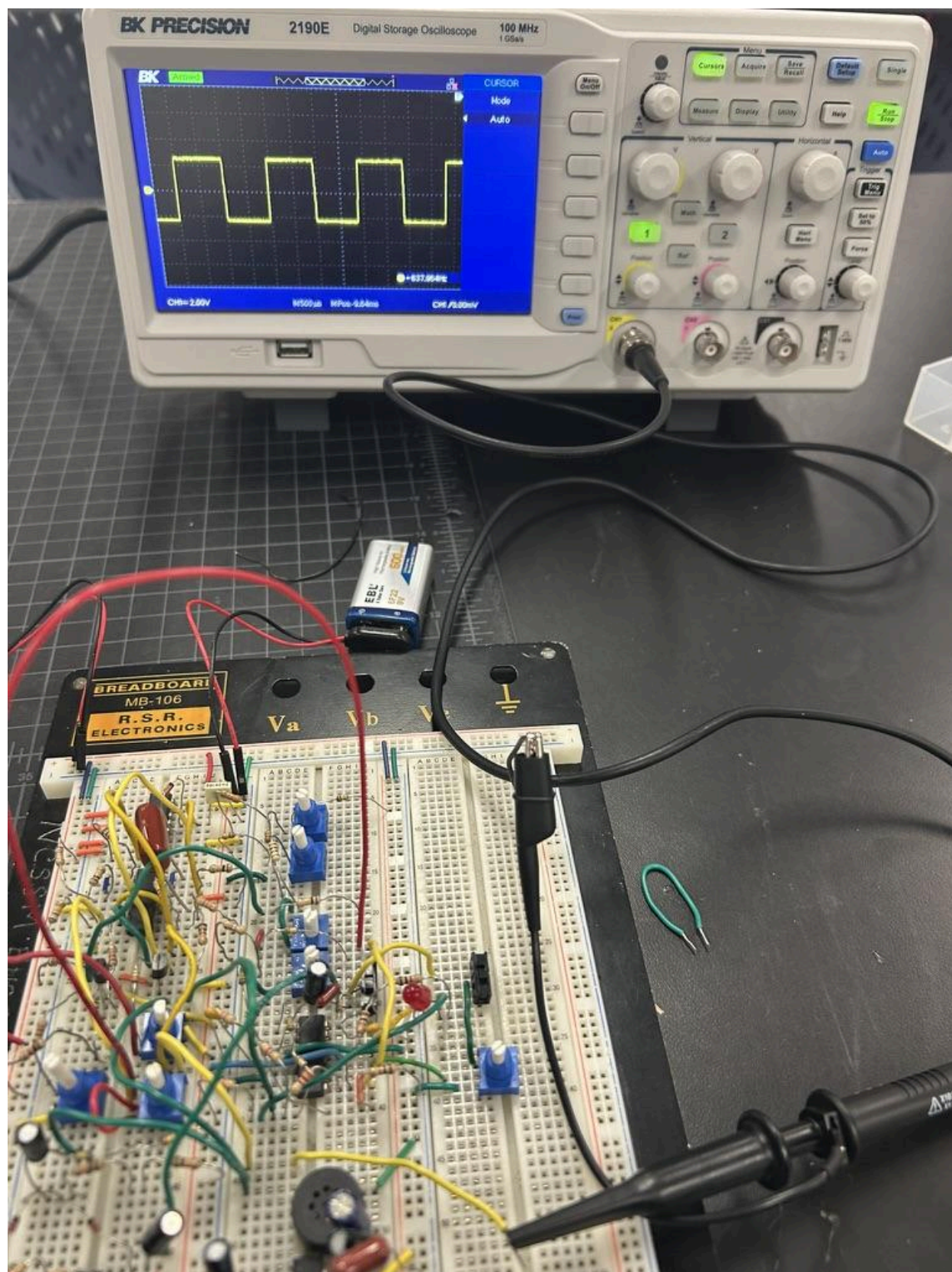


Low Frequency Oscillator



Photographs of System





[Video of Project](#)

Description of Design Evolution

At first, my design did not produce the correct sawtooth and pulse waves so I kept rebuilding it according to the schematic, and eventually, I was able to get the waves to work. Then, I noticed there was a lot of noise in the output. I tried numerous different methods to eliminate the noise, but the solution that worked was connecting all the unused pins of the Schmitt chip to ground, which completely got rid of the noise. I also did not build the low pass filter. Other than that, everything stayed true to the original design.

Parts List

Part	Quantity
Battery Clip	2
9V Battery	2
Breadboard	1
Switch	2
Speaker	1
1N4148 Diode	3
LED	1
CD40106 Schmitt Trigger Inverter Chip	1
TL074 Chip	2
LM386 Chip	1
TL084 Chip	1
10Ω Resistor	1
1kΩ Resistor	5

10k Ω Resistor	5
12k Resistor	1
20k Ω Resistor	1
33k Resistor	1
47k Ω Resistor	2
68k Ω Resistor	2
100k Ω Resistor	12
1M Ω Resistor	2
1.5k Ω Resistor	1
200k Ω Resistor	2
470 Ω Resistor	1
2.2k Ω Resistor	1
100k Ω Potentiometer	5
500k Ω Potentiometer	1
1M Ω Potentiometer	1
1k Ω Potentiometer	1
2.2nF Capacitor	1
1uF Capacitor	5
0.1uF Capacitor	3
10uF Capacitor	2
0.047uF Capacitor	1
220uF Capacitor	1

Reflection

I had a lot of trouble at first trying to get all the wave outputs to work correctly. This was because I didn't use the test points while building the circuit at first. However, after rebuilding the circuit several times, I finally got the pulse and sawtooth waves to work.

After that, the triangle wave was easier, and I didn't encounter any problems with the rest of the circuit. Next time, I would definitely use the test points before building the entire circuit. I would also try to implement modifications to change the sound, make the circuit more organized (such as making the potentiometers easier to access), and color-code the wires. I realized I had a hard time continuing to build the circuit the next day because I couldn't remember where I left off. In conclusion, although I faced several challenges, my project turned out fine, and now I know what I can improve on next time.